

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0024

Hilltop Alte 138 kV
July 26, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0024. The Interconnection Customer proposes a 323.7 MW wind facility interconnecting at the Hilltop Alte 138 kV substation in the New Jersey area. The request was received on April 11, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 20 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,339,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: W Orange Substation – Blanchard 138 KV line
Contingency: Aeropuerto – Hilltop Alte Substation 138 KV
Loading: 146.0% (Rating: 192 MVA)
DFAX: 100.0%
Pre-project loading: 22.8%
2. Facility: Blanchard – Hilltop Alte 138 KV
Contingency: Hilltop Alte – Aeropuerto 138 KV L
Loading: 142.2% (Rating: 213 MVA)
DFAX: 100.0%
Pre-project loading: 10.1%
3. Facility: Hilltop Alte – Aeropuerto 138 KV
Contingency: Blanchard – Hilltop Alte 138 KV
Loading: 135.5% (Rating: 223 MVA)
DFAX: 100.0%
Pre-project loading: 9.6%
4. Facility: Brown – Aeropuerto 138 KV
Contingency: Hilltop Alte – Blanchard 138 KV L
Loading: 119.5% (Rating: 235 MVA)
DFAX: 100.0%
Pre-project loading: 18.3%
5. Facility: West Orange – Magnolia 1708 138 KV
Contingency: Aeropuerto – Hilltop Alte 138 KV
Loading: 109.4% (Rating: 189 MVA)
DFAX: 69.6%
Pre-project loading: 9.5%

6. Facility: Locust Grove – Shredder 138 KV line
 Contingency: Pine Tree Acres – Crawfordsville Power Plant 138 KV L
 Loading: 104.4% (Rating: 241 MVA)
 DFAX: 11.5%
 Pre-project loading: 89.0%
7. Facility: Crawfordsville Power Plant – Pine Tree Acres 138 KV
 Contingency: Pipe – Rexnord 138 KV
 Loading: 103.5% (Rating: 363 MVA)
 DFAX: 8.2%
 Pre-project loading: 96.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Orange Substation – Blanchard 138 KV line	Rebuild line with higher-capacity conductor	\$1,205,000	100% (\$1,205,000)
Blanchard – Hilltop Alte Substation 138 KV L	Rebuild line with higher-capacity conductor	\$1,711,000	100% (\$1,711,000)
Hilltop Alte – Aeropuerto 138 KV	Rebuild line with higher-capacity conductor	\$1,449,000	100% (\$1,449,000)
Brown – Aeropuerto 138 KV	Rebuild line with higher-capacity conductor	\$763,000	100% (\$763,000)
West Orange – Magnolia 1708 138 KV	Reconductor line	\$378,000	100% (\$378,000)
Locust Grove – Shredder 138 KV line	Reconductor line	\$518,000	100% (\$518,000)
Crawfordsville Power Plant – Pine Tree Acres 138 KV	Reconductor line	\$315,000	100% (\$315,000)
	Total allocated to project		\$6,339,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.